Kanan Mahammadli

PhD Student, Max Planck Institute for Informatics & Software Systems — CS@Max Planck PhD Program
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Profile

PhD student in the CS@Max Planck program with a background in mathematics, statistics, representation learning, optimization, and generative modeling. My research examines how learned representations support reasoning and sequential decision-making, and why model performance degrades on long-horizon, compositional, and out-of-distribution tasks. I am broadly interested in foundation models and autonomous agents across language, multimodal, and embodied settings, with an emphasis on rigorous evaluation and methods for understanding and improving model behavior.

Research Interests

Foundation Models and Autonomous Agents, Representation Learning and Generative Modeling, Model Adaptation and Post-Training, Multimodal and Embodied Learning

Education

Saarland University — Saarbrücken, Germany
PhD in Computer Science, Preparatory Phase — CS@Max Planck Doctoral Program — *Sep. 2025 – Present*

Middle East Technical University — Ankara, Turkey
B.S. in Mathematics, Minor in Statistics — CGPA: 3.64/4.00 — *Sep. 2020 – Jun. 2025*

- Graduated with High Honors; recipient of a full-tuition scholarship for academic excellence.
- Relevant graduate coursework: Machine Learning, Deep Learning, Reinforcement Learning, Sequence Models in Multimedia.

Research Experience

Doctoral Researcher — Oct. 2025 – Jul. 2026
Max Planck Institute for Informatics — *Saarbrücken, Germany*

- Completed two Research Immersion Labs, supervised by Prof. Jan Eric Lenssen and Dr. Marc Habermann, on self-supervised latent-action representations for robot learning.
- Investigated continuous action representations learned from human videos and their use as task-relevant conditioning for downstream robot action generation.
- Evaluated learned representations through reconstruction, retrieval, and downstream manipulation experiments, combining representation learning with diffusion- and flow-matching-based action modeling.
- The project motivated broader research questions on representation sufficiency, long-horizon decision-making, compositional generalization, and interpretable failure analysis in learned autonomous systems.

Undergraduate Student Researcher — Jul. 2023 – Jun. 2025
AI Lab, METU BILTIR Center — Supervised by Prof. Seyda Ertekin — *Ankara, Turkey*

- Developed SLLMBO, a sequential hyperparameter-optimization framework that combines LLM-guided configuration proposals with Tree-structured Parzen Estimation.
- Evaluated four major LLMs across 14 classification and regression tasks; the proposed framework outperformed standard Bayesian-optimization baselines on 9 tasks.

Undergraduate Computer Vision Research Intern — Jul. 2022 – Sep. 2022
University of Houston — Supervised by Prof. Ioannis Kakadiaris — *Houston, TX, USA*

- Worked on human detection from UAV imagery captured under varying viewpoints, distances, and weather conditions, using feature pyramid networks for multi-scale visual recognition.
- Worked on face representation and clustering using FaceNet embeddings and FAISS for efficient similarity-based analysis.

Preprints

Mahammadli, K. and Ertekin, S. *Sequential Large Language Model-Based Hyperparameter Optimization.*
arXiv:2410.20302, 2024.

Mahammadli, K., Bereketoglu, A. B., and Kabakci, A. G. *Class-Specific Data Augmentation: Bridging the Imbalance in Multiclass Breast Cancer Classification.*
arXiv:2310.09981, 2023.

Selected Industry Experience

Senior Data and Optimization Scientist — Nov. 2023 – Jul. 2025
SmartKiwi AI — *London, UK (Remote)*

- Built an end-to-end decision-support platform for demand forecasting and inventory optimization, enabling users to obtain tailored forecasts and support operational decisions through an interactive interface.
- Developed and deployed a high-frequency passenger forecasting and optimization system at Istanbul Grand Airport for lane and staff allocation, achieving 92% forecasting accuracy, reducing wait times by 25–35%, and improving staff productivity.

Mid-level Data Scientist — May 2023 – Nov. 2023
SmartKiwi AI — *London, UK (Remote)*

- Designed a polygon-based region assignment strategy for Sendeo's courier operations, replacing street-based assignments with optimized service regions and improving delivery coverage and workload balance.
- Contributed to automated courier scheduling and route assignment systems, replacing manual planning with data-driven optimization across the delivery network.

Junior Data Scientist — Jan. 2023 – May 2023
SmartKiwi AI — *London, UK (Remote)*

- Developed a geospatial demand modeling strategy for car-sharing systems by partitioning cities into hexagonal grids, improving vehicle distribution and service availability across multiple deployments.
- Built a surge pricing model for more than 80 kitchen locations by analyzing user behavior and dynamically adjusting delivery fees, increasing monthly delivery revenue by 50%, while maintaining user-conversion rates.

Data Science Intern — Nov. 2021 – Apr. 2022
Affable AI — *Singapore (Remote)*

- Improved the sentiment analysis pipeline, reducing batch prediction latency from 180 seconds to 23 seconds and enabling faster large-scale NLP processing.
- Deployed a transformer-based brand classification model that improved influencer matching quality by 25% for digital marketing applications.

Technical Skills

Programming and ML: Python, PyTorch, NumPy, pandas, scikit-learn, Transformers
Tools and Systems: Git, Linux, Slurm, DeepSpeed, Weights & Biases
Methods: Representation Learning, Generative Modeling, Diffusion and Flow Matching, Hyperparameter Optimization, Model Evaluation